

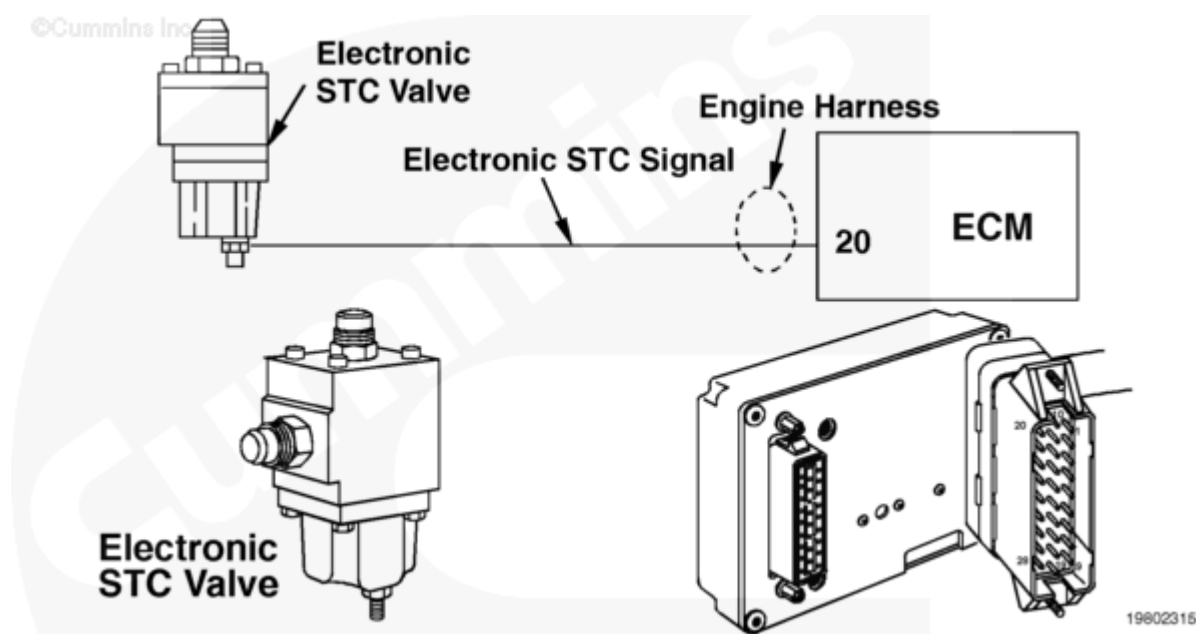
Fault Code: 113

Electronic Step Timing Control Circuit - Voltage Above Normal or Shorted to High Source

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Overview

Codes	Reason	Effect
Fault Code: 113 PID(P): S20 SPN: FMI: 3 Lamp: On SRT: 00-619	High voltage detected at the electronic step timing control circuit.	Possible reduced engine performance.



Electronic Step Timing Control (STC) Circuit

Circuit Description

The electronic step timing control actuator controls the step timing control valve on the fuel pump. The electronic step timing control actuator has a single circuit; step timing control driver/torque output driver. The Engine Control Module (ECM) adjusts the engine timing from advance to normal with the electronic step timing control actuator.

Component Location

The electronic step timing control valve is mounted on the left side of the engine.

Conditions For Running The Diagnostics

This diagnostic runs continuously when the engine is running.

Conditions For Setting The Fault Codes

The Engine Control Module (ECM) detected the electronic step timing control voltage was out of range high.

Action Taken When The Fault Code Is Active

- The ECM illuminates the amber CHECK ENGINE lamp immediately when the diagnostic runs and fails.
- The electronic step timing control valve is in full timing advance.

For Power Generation Applications:

- The generator set controller displays the fault immediately when the diagnostics runs and fails.
- The electronic step timing control valve is in full timing advance.

Conditions For Clearing The Fault Code

- To validate the repair, perform a key cycle, start the engine and let it idle for 1 minute.
- The fault code status displayed by the recommended Cummins® electronic service tool or equivalent will change to INACTIVE immediately after the diagnostic runs and passes.
- The ECM will turn off the amber CHECK ENGINE lamp immediately after the diagnostic runs and passes.
- The “Reset All Faults” command in the recommended Cummins® electronic service tool or equivalent can be used to clear active and inactive faults.

For Power Generation Applications:

- To validate the repair, start the engine and let it run for 1 minute at no load.
- The generator set controller will turn off the indicator immediately after the user presses reset.
- The “Reset All Faults” command in the recommended Cummins® electronic service tool or equivalent can be used to clear active faults.

Shoptalk

Possible causes of this fault code include:

- Malfunctioning or damaged engine wiring harness.
- Malfunctioning or damaged electronic step timing control actuator.
- Loose or damaged connectors.

Refer to Troubleshooting Fault Code t05-113 (</qs3/pubsys2/xml/en/procedures/98/98-t05-113.html>)

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